ALL RTU EXAM QUESTIONS LAST 4 YEARS (UNIT 3, 4, 5)

INFORMATION THEORY CODING (V SEM CSE)

QUESTION BANK (UNIT 3, 4, 5)

PART A

Q1: Differentiate between random error & burst error. (RTU 2024)

Q2: Classify the linear block code. (RTU 2024)

Q3: Discuss the cyclic code and its generation. (RTU 2024)

Q4: Explain the 2 properties of Galois Field. (RTU 2024)

Q5: Compare convolutional code over coding methods. (RTU 2024)

Q6: Describe the differences between state diagram and trellis diagram of convolution code. (RTU 2024)

Q7: What do you mean by Galois Field. (RTU 2023)

Q8: What is the property to satisfied by the linear block code. (RTU 2023)

PART B

Q1: Illustrate the concept of systematic and non-systematic code. (RTU 2024)

Q2: The following polynomial f(x) and g(x) are defined over GF(3). (RTU 2024)

$f(x)= 2+x+x^3+2x^4$

$g(x)= 1+2x^2+2x^4+x^5$

calculate the addition and multiplication of above 2 polynomials. (RTU 2024)

Q3: Describe viterbi algorithm & maximum likelihood decoding in detail. (RTU 2024, 2022)

Q4: What is the standard array in syndrome decoding? Explain with suitable example. (RTU 2023)

Q5: The parity matrix (sub matrix) for (6,3) linear block code is given by (RTU 2023)

$$P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

(a)Find generator matrix and parity check matrix

(b)Find all code word

Q6: For systematic LBC(6,3), the three parity check digit given as below: (RTU 2022)

$C_4=d_1+d_2+d_3$

$C_5=d_1+d_2$

$C_6=d_1+d_3$

(i)Construct generator matrix

(ii)Construct all code vectors

Q7: A generator polynomial of (7,4) cyclic code is given $G(p)=p^3+p+1$. Find all code vectors and generator matrix for the code in systematic form of cyclic code. (RTU 2022)

PART C

Q1: Consider a (7,4) linear block code with the parity check matrix H given by (RTU 2024)

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(a)Contruct all codeword

(b)Show that this code is hamming code

Q2: The generator polynomial of a (6,3) cyclic code is $g(x)=1+x^2$ (RTU 2024)

(a)Find all the codeword for this code

(b)How many errors can this code correct?

Q3: Write short notes on (RTU 2024)

(a)Sequential decoding

(b)Transfer function of convolutional Code

Q4: The parity bits of (7,4) Linear systematic block code are generated by (RTU 2023)

$C_5=d_1+d_3+d_4$

$C_6=d_1+d_2+d_3$

$C_7=d_2+d_3+d_4$

Find the generator matrix G & parity check matrix H for this code. Draw the encoder circuit.

Q5: A channel encoder uses a (7, 4) linear systematic cyclic code & its T generator polynomial

X^3+x+1 is given: Determine the correct codevector transmitted, if the received code vector is: (RTU 2023)

i) 1011011
ii) 1101111

Q5: Discuss the procedure for generating systematic cyclic code. Draw and explain the systematic cyclic encoder for a (15,9) cyclic code with generator polynomial: (RTU 2023)

$G(x)= 1+x^3+x^4+x^5+x^6$

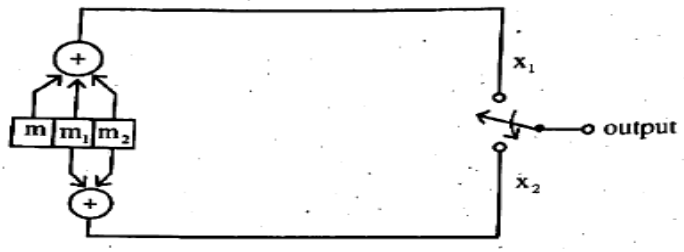
Q6: Explain the decoder of cyclic code with the help of block diagram. (RTU 2023)

Q7: For a systematic linear block encoder, the generator matrix is given by (RTU 2022)

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Decode the received code vector 1011101 and if it incorrect then recover the correct codeword.

Q8: A convolutional encoder with constraint length 3 rate ½ shown below: (RTU 2022)



Find out:

(a)Code rate

(b)Dimension of encoder

(c)Constraint length

(d)output sequence for the sequence message m= 10011

Q9: Explain the need of error correcting codes. How its encoding/ Decoding takes place? Explain with the help of parity example. (RTU 2021)

Q10: Write Short notes: (RTU 2021)

i) Hamming code and their applications
ii) Advantages and disadvantages of conlutional code